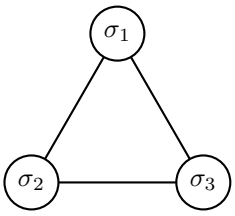
**Computational Physics I 3630/5630 Fall 2026**

**Homework 5**

**Due: Wednesday, October 21, 2026**

1. **Three-spin Ising model and the transition matrix**. Consider three Ising spins $\sigma_i = \pm 1$ arranged on a ring, with periodic boundary conditions ($\sigma_4 = \sigma_1$):




A configuration is $X = (\sigma_1, \sigma_2, \sigma_3)$, and its energy is

$$E(X) = -J(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1) - H(\sigma_1 + \sigma_2 + \sigma_3).$$

There are only $2^3 = 8$ possible configurations. We will denote them by

$$(+++),\ (++-),\ (+-+),\ (-++),\ (+--),\ (-+-),\ (--+),\ (---),$$

where, for example, $(+-+)$ means $(\sigma_1, \sigma_2, \sigma_3) = (+1, -1, +1)$. This small system is a useful example in which the full Markov chain can be constructed explicitly. For the numerical results below, take $H = 0$.

**(a) Transition graph** [35%]. Using the standard single-spin Metropolis algorithm, construct the complete transition graph among the eight configurations. In one Monte Carlo step, choose one of the three spins with equal probability, propose flipping it, and accept or reject the proposed move according to the Metropolis rule. Label every directed edge $X \to Y$ with its transition probability $P(X \to Y)$. Your graph should also account for the probability of remaining in the same configuration when a proposed move is rejected.

**(b) Transition matrix and stationary distribution** [30%]. Using the ordering of states given above, construct the full $8 \times 8$ transition matrix $\mathbf{P}$, using the right-vector convention $\boldsymbol{\pi}_{n+1} = \mathbf{P}\boldsymbol{\pi}_n$. Numerically diagonalize $\mathbf{P}$ and identify the eigenvector corresponding to its largest eigenvalue. Normalize this eigenvector so that its components sum to one; this gives the stationary probability distribution $\boldsymbol{\pi}$.

Make bar histograms of the eight probabilities

$$\pi(+++),\ \pi(++-),\ \pi(+-+),\ \pi(-++),\ \pi(+--),\ \pi(-+-),\ \pi(--+),\ \pi(---)$$

at three temperatures

$$T = 0.1\, J/k_B, \qquad T = J/k_B, \qquad T = 10\, J/k_B.$$

Describe the distribution you obtain and explain how it changes with temperature.

**(c) Relaxation time** [35%]. From the eigenspectrum of $\mathbf{P}$, identify the eigenvalue $\lambda_2$ with the second-largest magnitude, after the stationary eigenvalue $\lambda_1 = 1$. Taking one Monte Carlo step as the unit of time, $\delta t = 1$, compute

$$\tau_{\text{relax}} = -\frac{1}{\ln|\lambda_2|}$$

and plot $\tau_{\text{relax}}$ as a function of temperature. Describe the temperature dependence and explain qualitatively why relaxation becomes slow in the regime where you observe a large $\tau_{\text{relax}}$.

2. **Monte Carlo simulation of the 1D Ising model** [100%]. Write a code to implement a Markov-chain Monte Carlo (MCMC) simulation of the 1D Ising model and compare the simulation results with the analytical

expressions provided on page 3 (Problem 3). PHYS 3630 students may use these formulas directly, without deriving them. A configuration of a linear chain is specified by a set of Ising variables

$$X = \{\sigma_1, \sigma_2, \ldots, \sigma_N\}, \qquad \sigma_i = \pm 1,$$

where $\sigma_i$ denotes the spin at site $i$. The total energy and magnetization are

$$E(X) = -J \sum_{i=1}^{N} \sigma_i \sigma_{i+1} - H \sum_{i=1}^{N} \sigma_i, \qquad M(X) = \sum_{i=1}^{N} \sigma_i. \tag{1}$$

We assume periodic boundary conditions, $\sigma_{N+1} = \sigma_1$.

The MCMC simulation generates a sequence of configurations,

$$X_1 \to X_2 \to X_3 \to \cdots \to X_{N_{\text{data}}},$$

from which you obtain the corresponding energies $E(X_1), E(X_2), \ldots$ and magnetizations $M(X_1), M(X_2), \ldots$. After thermalization, use these sampled configurations to estimate the energy density, heat capacity per site, magnetization density, and magnetic susceptibility per site:

$$\varepsilon = \frac{\langle E \rangle}{N}, \qquad c = \frac{\langle E^2 \rangle - \langle E \rangle^2}{N k_B T^2}, \qquad m = \frac{\langle M \rangle}{N}, \qquad \chi = \frac{\langle M^2 \rangle - \langle M \rangle^2}{N k_B T}. \tag{2}$$

Here the brackets denote Monte Carlo averages; for example,

$$\langle E \rangle = \frac{1}{N_{\text{data}}} \sum_{n=1}^{N_{\text{data}}} E(X_n), \qquad \langle E^2 \rangle = \frac{1}{N_{\text{data}}} \sum_{n=1}^{N_{\text{data}}} E(X_n)^2,$$

and similarly for $\langle M \rangle$ and $\langle M^2 \rangle$.

Plot the temperature dependence of $\varepsilon$, $c$, $m$, and $\chi$ from your simulations. Compare your numerical results with the exact analytical formulas provided on page 3 in Problem 3, including the zero-field eigenvalues given there. These formulas may be used directly by PHYS 3630 students.

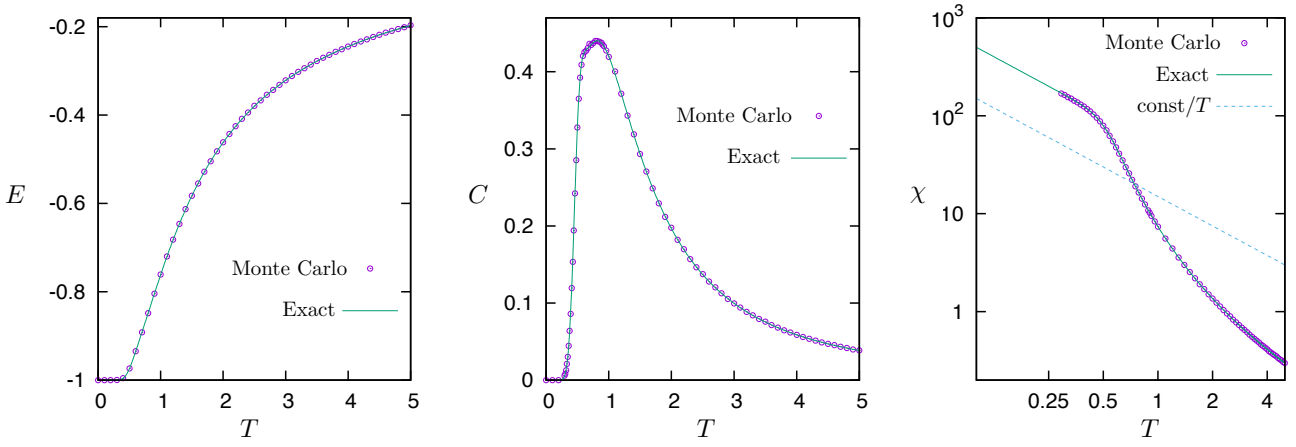



Figure 1: Example MCMC results for a 1D Ising chain with periodic boundary conditions, $J = 1$, $H = 0$, and $N = 50$, compared with exact transfer-matrix results. From left to right: energy per site, heat capacity per site, and magnetic susceptibility per site as functions of temperature.

3. **Transfer-matrix method for the 1D Ising model.**

*This problem is required for PHYS 5630 and is bonus credit for PHYS 3630. The transfer-matrix method is more advanced than the material we have covered so far; the goal is to introduce an exact solution of the 1D Ising model that can be used to benchmark the Monte Carlo results in Problem 2.*

The 1D Ising chain with periodic boundary conditions can be solved exactly using the transfer-matrix method. Its partition function is

$$Z_N(\beta J, \beta H) = \sum_X e^{-\beta E(X)} = \sum_{\{\sigma_i\}} \exp\left[\beta J \sum_{i=1}^{N} \sigma_i \sigma_{i+1} + \beta H \sum_{i=1}^{N} \frac{\sigma_i + \sigma_{i+1}}{2}\right]. \tag{3}$$

where

$$\beta = \frac{1}{k_B T}, \qquad \sum_{\{\sigma_i\}} = \sum_{\sigma_1=\pm 1} \sum_{\sigma_2=\pm 1} \cdots \sum_{\sigma_N=\pm 1}.$$

Show that the Boltzmann factor can be written as a product of matrix elements, so that the partition function becomes

$$Z_N(\beta J, \beta H) \;=\; \sum_{\{\sigma_i\}} \mathbb{T}_{\sigma_1,\sigma_2}\, \mathbb{T}_{\sigma_2,\sigma_3} \cdots \mathbb{T}_{\sigma_{N-1},\sigma_N}\, \mathbb{T}_{\sigma_N,\sigma_1} = \mathrm{Tr}\,(\mathbb{T}^N). \tag{4}$$

The $2 \times 2$ matrix $\mathbb{T}$ is the *transfer matrix*; its elements depend on $T$, $J$, and $H$. Obtain the explicit form of $\mathbb{T}$ [30%].

The transfer matrix can be diagonalized. Denote its two eigenvalues by $\lambda_+$ and $\lambda_-$. Find these eigenvalues [15%] (for general nonzero $H$), and then show that [5%]

$$Z_N(\beta J, \beta H) = \lambda_+^N + \lambda_-^N. \tag{5}$$

Thermodynamic quantities can be obtained from derivatives of the partition function:

$$\varepsilon \;=\; -\frac{1}{N}\frac{\partial \ln Z_N}{\partial \beta}, \qquad c = \frac{k_B \beta^2}{N}\frac{\partial^2 \ln Z_N}{\partial \beta^2}, \tag{6}$$

$$m \;=\; \frac{1}{N\beta}\frac{\partial \ln Z_N}{\partial H}, \qquad \chi = \frac{1}{N\beta}\frac{\partial^2 \ln Z_N}{\partial H^2}. \tag{7}$$

For $H = 0$, the two transfer-matrix eigenvalues have the simple expressions

$$\lambda_+ = e^{\beta J} + e^{-\beta J}, \qquad \lambda_- = e^{\beta J} - e^{-\beta J}. \tag{8}$$

Using these eigenvalues, the following analytical expressions hold at $H = 0$. PHYS 3630 students may use these results directly to plot the analytical curves for comparison with the Monte Carlo results in Problem 2.

[Extra 30% credit: derive these formulas yourself.]

$$\varepsilon \;=\; -J\left(\lambda_+^{N-1}\lambda_- + \lambda_-^{N-1}\lambda_+\right)/Z, \tag{9}$$

$$c \;=\; 4k_B(\beta J)^2\left[\lambda_+^{2N-2} - \lambda_-^{2N-2} + 4(N-1)(\lambda_+\lambda_-)^{N-2}\right]/Z^2, \tag{10}$$

$$\chi \;=\; \beta\left[\lambda_+^{N-1}\left(e^{\beta J} + e^{3\beta J}\right) + \lambda_-^{N-1}\left(e^{\beta J} - e^{3\beta J}\right)\right]/Z. \tag{11}$$

Here $Z = Z_N = \lambda_+^N + \lambda_-^N$, and $m = 0$ in the zero-field limit. Plot the temperature dependence of $\varepsilon$, $c$, and $\chi$ using these formulas and compare them with your numerical results from Problem 2. These comparisons are part of the 100% assigned to Problem 2.